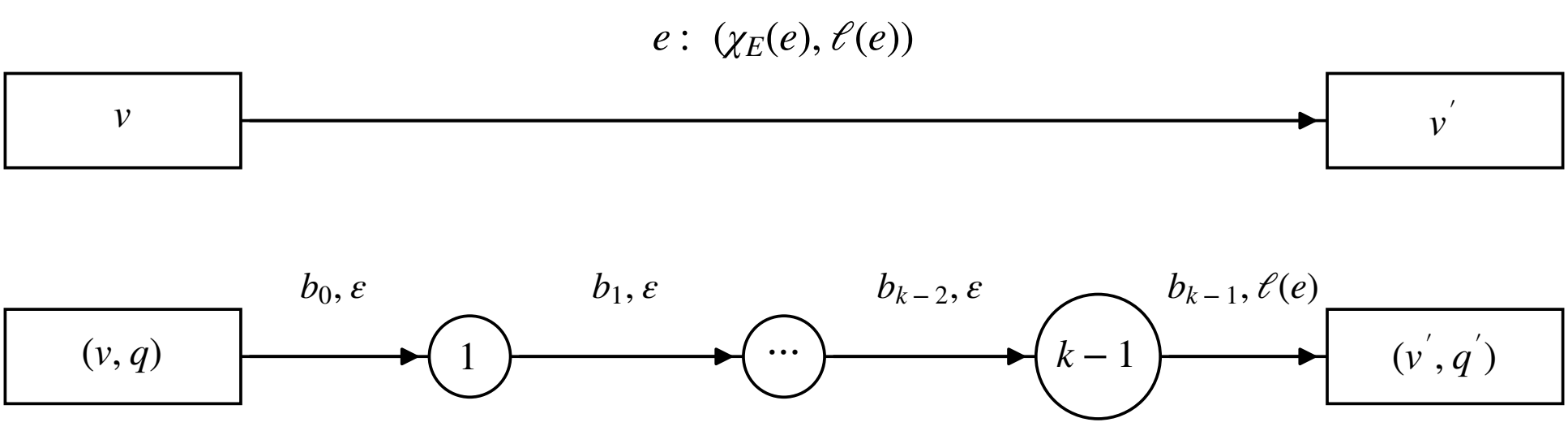


$$q' = \delta(q, \eta(e)); b_0 \cdots b_{k-1} = \lambda(q, \eta(e)) \neq \varepsilon$$